



Abundance and diversity of termites in a savanna woodland reserve in tropical Australia

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Abstract

This study contributes to the paucity of knowledge on the structure of Australian tropical savanna termite assemblages, compared with that available for tropical savannas elsewhere in the world. Termite species diversity, frequency of occurrence and functional diversity in Charles Darwin National Park, a savanna woodland reserve in tropical northern Australia, were assessed. In four 50 × 50 m sites, a direct search and bait sampling protocol was used to sample mounds, standing and lying dead wood, litter, soil and arboreal nest microhabitats for termites between April and June 2003. Results were compared with *a priori* predictions of termite community structure based on the work by Braithwaite *et al.* (1988). Sixteen species from nine genera and three families (Mastotermitidae, Rhinotermitidae and Termitidae) were recorded, including the first record of *Ephelotermes taylori* (Hill) from open savanna woodland habitat and the first official record of *Macrognathotermes errator* Miller from north Australian savanna. The family Termitidae represented 69% of termite occurrences. From this family, the wood/soil interface feeder *Nasutitermes eucalypti* (Mjöberg) represented 30% of encounters. High levels of termite activity were recorded at baits within 6 weeks. Four nesting habits were represented: epigeal (mounds above ground), hypogeal (subterranean nests), nests within wood and arboreal nests. Termites were most frequently encountered in epigeal mounds and lying dead wood. Wood and wood/soil interface feeders were the most common. The findings partially supported the *a priori* predictions proposed. The termite assemblage structure was represented by species commonly found in other Northern Territory savanna woodland habitats. However, species diversity was lower; soil feeders and live wood feeders were less common and grass-harvesting species were absent. Soil nitrogen and phosphorus and species richness and patchiness of ground vegetation may contribute to these differences. The lower species diversity recorded contrasts with the richer termite faunas of tropical savannas recorded elsewhere.

Key words functional groups, Northern Territory, termite assemblage structure, termite sampling, wood feeders.

INTRODUCTION

Termites dominate macroinvertebrate abundance and biomass in tropical soils, where they are major ecosystem engineers (Lavelle *et al.* 1997; Bignell & Eggleton 2000; Jones 2000). Through their diverse feeding and nesting activities they are key mediators of pedological and ecological processes (Holt & Lepage 2000; Lavelle *et al.* 2001; Dawes-Gromadzki & Spain 2003). In tropical Australia, termites promote soil primary productivity through their influence on nutrient cycling, bioturbation and soil hydraulic conductivity (Lee & Wood 1971; Holt & Coventry 1990; Holt *et al.* 1996; Holt & Lepage 2000). These effects can also influence the distribution and growth of vegetation (Spain & Okello-Oloya 1985; Spain & McIvor 1988) and communities of micro-organisms and other biota (Abensperg-Traun & De Boer 1990; Spain *et al.* 2004).

Despite the recognised importance of termites to ecosystem function in tropical north Australia, there has been little effort devoted to the documentation of termite community inventories (Braithwaite *et al.* 1988; Dawes-Gromadzki 2005a). At a given site the composition of a local termite assemblage will govern the overall impacts of termites on ecosystem processes (Lavelle *et al.* 1997; Jones & Prasetyo 2002). This is because most termite assemblages are made up of species that represent a diverse range of feeding (e.g. wood, soil, grass) and nesting (e.g. mounds, hypogeal and arboreal nests, nests in wood) strategies (i.e. different 'functional groups'). Consequently different species and groups of species are likely to have different pedological and ecological effects on ecosystem processes such as bioturbation, soil formation, nutrient cycling, soil water infiltration and storage (Jones & Prasetyo 2002; Dawes-Gromadzki 2005b).

Detailed knowledge of local species diversity as well as the diversity of functional groups of termites (different groups of species that have different ecological strategies, e.g. different

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feeding habits) is critical. It is a prerequisite for evaluating and quantifying the type and extent of their functional roles in ecosystem processes, i.e. the contributions that different termite species and functional groups make to the regulation of different ecosystem processes. It is also important for the development of sound management practices, e.g. agricultural, grazing and fire management (Abensperg-Traun 1991; Inoue *et al.* 2001; Dawes-Gromadzki 2005b; Dawes-Gromadzki 2007).

This paper presents data on the diversity of termite species (number of different species recorded) and functional groups (number of different termite functional groups based on feeding and nesting strategies) and the relative abundance of termites (number of encounters of each species as proportion of total) in a savanna woodland habitat of Charles Darwin National Park (CDNP) in the Northern Territory (NT), Australia. In this paper I seek to test the generality of observations made from similar habitats in Australia, where savanna woodland habitats are species-rich compared with monsoon rainforest. Due to the limited nature of such studies, comparisons will mostly rely on the work of Braithwaite *et al.* (1988).

Based on the work of Braithwaite *et al.* (1988), my predictions for the termite community structure of CDNP are a termite assemblage characterised by fairly high species diversity (30+ species) where wood- (*Mastotermes darwiniensis*, *Coptotermes acinaciformis*, *Schedorhinotermes* and *Microcerotermes* spp.) and soil-feeding (*Termes* spp.) termites are dominant, grass-feeding species (*Amitermes*, *Tumulitermes* and *Nasutitermes* spp.) are common, and nesting habits are predominantly hypogeal (subterranean nests) or in epigeal (surface) mounds. I also compare the termite assemblage of CDNP with those overseas, and discuss the biogeography of the termite fauna surveyed.

METHODS

Study site

The study was conducted at CDNP, situated 5.5 km east of Darwin (12°27'S, 130°50'E) in the NT, Australia. The park vegetation is representative of typical northern Australian tropical savanna woodland. *Eucalyptus tetradonta* and *E. miniata* dominate the overstorey, over a patchy understorey of annual and perennial grasses. The park is of natural and cultural significance, protecting part of the nationally significant Port Darwin wetland (48 km²). The climate is wet-dry tropical and characterised by high temperatures throughout the year. Mean daily maximum and minimum temperatures are 31.9°C and 23.2°C, respectively (Bureau of Meteorology, NT). Mean annual rainfall is ~1700 mm, with over 90% falling during the wet season (November to April).

Sampling

Termite sampling was conducted between April and June 2003, the transitional period between the wet and dry season

when termite activity is known to be the highest (Dawes-Gromadzki & Spain 2003). Termites were quantitatively sampled using direct search and baiting techniques.

Direct searches

I randomly selected four 50 × 50 m sites and a standardised direct search sampling protocol based on searching different microhabitats for termites was followed in each (see Dawes-Gromadzki 2005a). This involves four trained people searched each site for 1 h (a total of 4 person-hours per site). Searching was conducted in pairs, with the two pairs starting at opposite site corners. Working back-and-forth across the site, the first member of each pair examined epigeal and intermediate mounds (new and old so that both the mound builders and secondary occupants were recorded) and foraging trails (carton runways) on trees and surface litter, including those from arboreal nests. The second member examined standing dead wood (including tree stumps), lying dead wood (including logs, branches and twigs), carton sheeting on dead wood and soil beneath rotten wood for termites. For each termite population encountered, representatives of the soldier caste were collected and preserved in 80% ethanol.

Termite species were also classified into nesting groups, and the nesting habit was recorded for each termite collection. The nesting groups were defined as: (1) hypogeal nesting (found below ground); (2) wood nesting (found in woody items); (3) epigeal (in mounds protruding above ground); and (4) arboreal (in tree nests) (Eggleton *et al.* 2002).

Baits

I selected two sites at random for additional bait sampling. A composite sampling protocol of soaked toilet paper rolls (500-sheet single-ply, unscented, unbleached) and stakes (40 × 35 × 300 mm seasoned, untreated, mountain ash *Eucalyptus regnans*) was used to maximise the species diversity sampled (see Dawes-Gromadzki 2003a,b). Paper and stake baits are interchanged, with six baits placed at 5 m spacing, along each of two parallel transect lines, located 20 m apart and placed symmetrically in the site. Paper rolls were reinforced at each end with tape and buried to 30 mm. Paper baits were identified by flagging tape, threaded through each roll and laid on the ground surface after burial. Stakes were buried vertically to 250 mm. Paper and stake baits were pre-soaked in water for 1 min and 48 h, respectively. At 3, 6 and 10 weeks baits were extracted and inspected for termites. Soldiers were collected and baits were placed back in their original position and orientation.

Species were also assigned to feeding groups based on literature available on Australian termites and *in situ* observations (Braithwaite *et al.* 1988; Spain & Reddell 1996; Tayasu *et al.* 1998; Dawes-Gromadzki & Spain 2003; Dawes-Gromadzki 2005a). The feeding groups were: (1) soil feeders (humus and mineral soil); (2) wood/soil interface feeders (extremely decayed wood that has lost its structure and is friable and soil-like); (3) wood feeders (live wood and/or

Table 1 Percentage occurrence, nesting and feeding habits and microhabitat locations of termite species collected in Charles Darwin National Park near Darwin between April and June 2003. The checklist represents all species recorded from direct search and bait sampling

Species	Direct search (n = 80)	Baits (n = 16)	Total (n = 96)	Feeding group†	Nesting group‡	Microhabitat found§
Mastotermitidae						
<i>Mastotermes darwiniensis</i> Froggatt	3.8	0	3.1	W	W	LDW
Rhinotermitidae						
<i>Coptotermes acinaciformis</i> Froggatt	7.5	0	6.3	W	M	M, SDW, LDW
<i>Schedorhinotermes actuosus</i> (Hill)	12.5	25.0	14.6	W	H	LDW, SDW, M, B
<i>Heterotermes vagus</i> (Hill)	0	25.0	4.2	W	H	B
<i>Heterotermes venustus</i> (Hill)	0	18.75	3.1	W	H	B
Termitidae – Termitinae						
<i>Amitermes perelegans</i> (Hill)	1.3	0	1.0	S	M(O)	M
<i>Ephelotermes melachoma</i> Miller	1.3	0	1.0	W/S	M	M
<i>Ephelotermes taylora</i> (Hill)	6.3	0	5.2	W/S	H, M(O)	LDW, SDW, M
<i>Macrognathotermes errator</i> Miller	1.3	0	1.0	S	M(O)	LDW
<i>Macrognathotermes sunteri</i> (Hill)	2.5	0	2.1	W/S	M(O)	M
<i>Microcerotermes boreus</i> Hill	12.5	25.0	14.6	W	M	M, LDW, SDW, B
<i>Microcerotermes serratus</i> (Froggatt)	1.3	6.25	2.1	W	M	SDW, B
<i>Microcerotermes nervosus</i> Hill	7.5	0	6.3	W	M	M
Termitidae – Nasutitermitinae						
<i>Nasutitermes eucalypti</i> (Mjöberg)	36.3	0	30.2	W/S	M(O)	M, LDW, SDW
<i>Nasutitermes graveolus</i> (Hill)	1.3	0	1.0	W	A	LDW
<i>Nasutitermes longipennis</i> (Hill)	5.0	0	4.2	P	M	M, SDW

†P, polyphagous; S, soil; W, wood; W/S, wood/soil interface. ‡A, arboreal; H, hypogeal; M, mound; M(O), found in mounds of other termites; W, wood. §B, bait; LDW, lying dead wood; M, mound; SDW, standing dead wood.

sound to partially decayed wood excluding extremely decayed and friable wood); (4) litter feeders (leaf-litter and small wood items); (5) grass harvesters (living and dead grass); and (6) polyphagous feeders (opportunists on a range of woody and herbaceous materials).

Analysis

Termites were identified to species using keys and descriptions from Hill (1942) and Miller (1991), and by comparisons with type specimens from the Australian National Insect Collection (ANIC), CSIRO Division of Entomology, Canberra. Voucher specimens of all species are held at the CSIRO Tropical Ecosystems Research Centre, Darwin.

The sampling effort allowed a species inventory to be compiled and a semi-quantitative assessment of species relative abundance based on the number of encounters for each species in each site to be calculated. These data were used to provide an overall indication of termite community structure based on the relative frequency of occurrence of different species and functional groups (Jones 2000; Dawes-Gromadzki 2005a).

RESULTS

Species composition, distribution and relative abundances

Sixteen species representing nine genera and three families (Mastotermitidae, Rhinotermitidae and Termitidae) were identified, from 112 termite collections (Table 1). Species

were not identified for 16 samples, as no soldiers were present. Species from the family Termitidae constituted 69% of the total number of species sampled. *Nasutitermes eucalypti* was the most frequently recorded species, representing 30% of all termite colonies encountered. *Schedorhinotermes actuosus* and *Microcerotermes boreus* each comprised approximately 15% of the total termite assemblage (Fig. 1). Termites were most commonly encountered in mounds, followed by dead wood lying on the soil surface. Only three termite samples were collected from carton runways, none of which contained soldiers (Fig. 2).

Direct searches

The direct searches produced 93 (80 with soldiers) of the 112 termite collections, including 14 of the 16 termite species sampled. *Nasutitermes eucalypti* accounted for 36% of the samples (Table 1). Five species, *Amitermes perelegans*, *Ephelotermes melachoma*, *Macrognathotermes errator*, *Microcerotermes serratus* and *Nasutitermes graveolus*, were single encounters. The number of species sampled varied across microhabitats. Mounds (10 species) and lying dead wood (eight species) collectively yielded 13 of the 14 species sampled by direct searches. The remaining species, *M. serratus*, was recorded from standing dead wood, which recorded a total of seven species.

Baits

Baiting yielded 19 termite samples (16 with soldiers) from 72 bait inspections (24 baits each on three occasions). Five

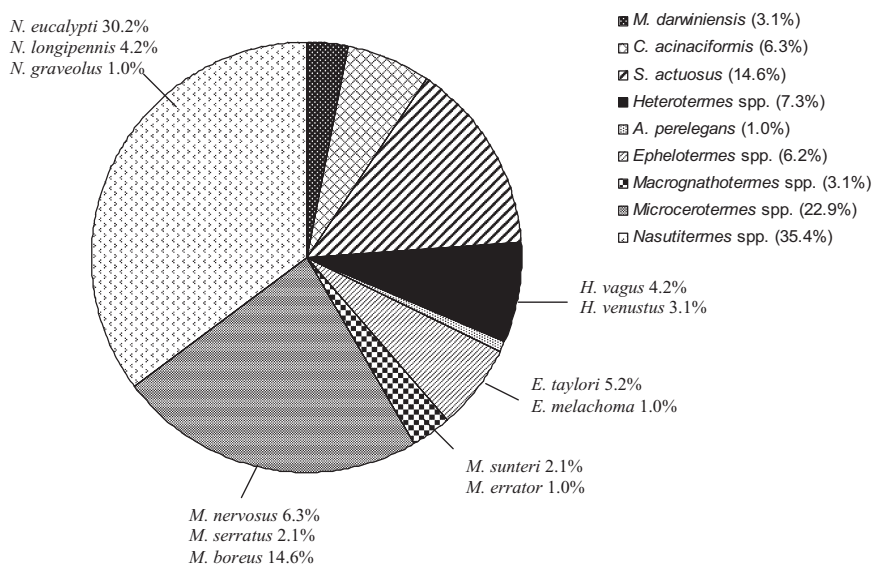


Fig. 1. Relative abundance of termite species as a proportion of total termite encounters at Charles Darwin National Park, Darwin, Northern Territory.

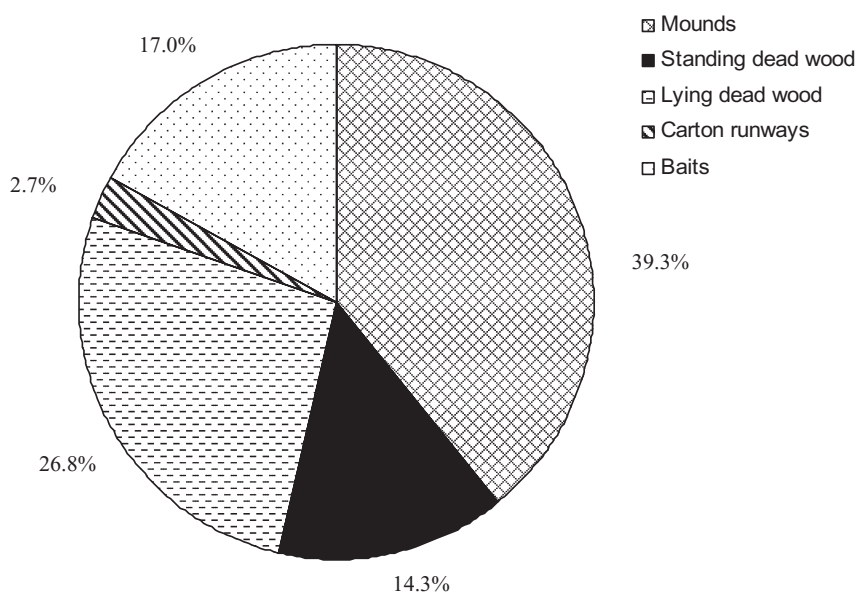


Fig. 2. Relative frequencies of occurrence of termites in different microhabitats from direct searches and bait sampling in Charles Darwin National Park, Darwin, Northern Territory.

species were identified: *S. actuosus*, *M. boreus*, *Heterotermes vagus*, *Heterotermes venustus* and *M. serratus*. The genus *Heterotermes* was only recorded from baits (Table 1). Of the 24 baits installed, 50% were attacked after 2.5 months. Stakes (58%) and paper (42%) baits were attacked at similar frequency.

No single species dominated baits. The frequency of occurrence of four species (*S. actuosus*, *M. boreus*, *H. vagus* and *H. venustus*) was similar, ranging from 19% to 25% of total bait observations (Table 1). These species attacked both bait types and were recorded in both sites. For both sites, 50% of the baits installed were attacked by week 10.

Feeding and nesting habits

The termite assemblage comprised four feeding groups (Fig. 3). Wood- and wood/soil interface-feeding species domi-

nated, comprising 56% and 25% of all species. The remaining species were either soil or polyphagous feeders.

Four nesting habits were represented, with 88% of species nesting either in epigeal mounds or below ground (Fig. 4). Epigeal-nesting species included *N. eucalypti*, *Nasutitermes longipennis*, *Microcerotermes* spp., *C. acinaciformis*, *Ephelotermes* spp., *Macrognathotermes* spp. and *A. perel-*

DISCUSSION

Termite species diversity

Overall, the CDNP survey yielded 16 termite species from nine genera and three families. The termite assemblage as a

Fig. 3. Proportions of different termite feeding groups represented across the 16 termite species collected from direct search and bait sampling in Charles Darwin National Park, Darwin, Northern Territory. Note: litter- and grass-feeding termite species were absent.

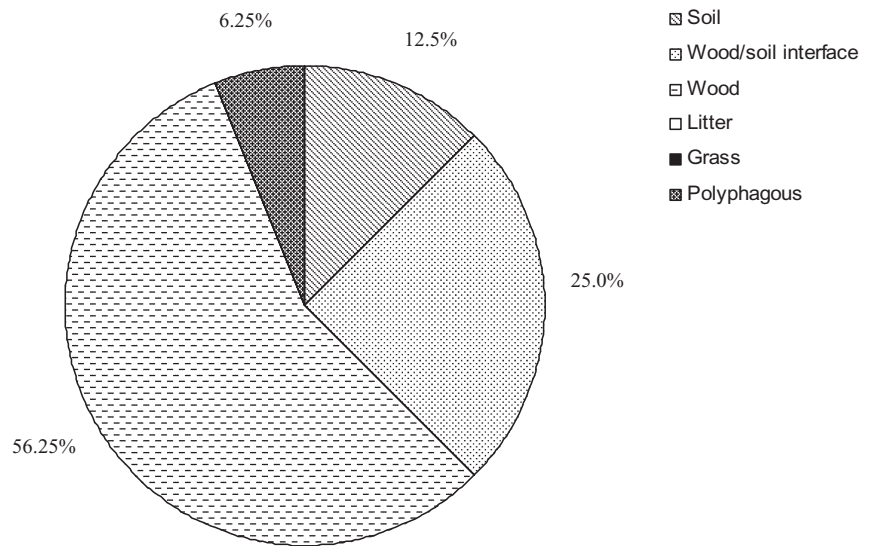
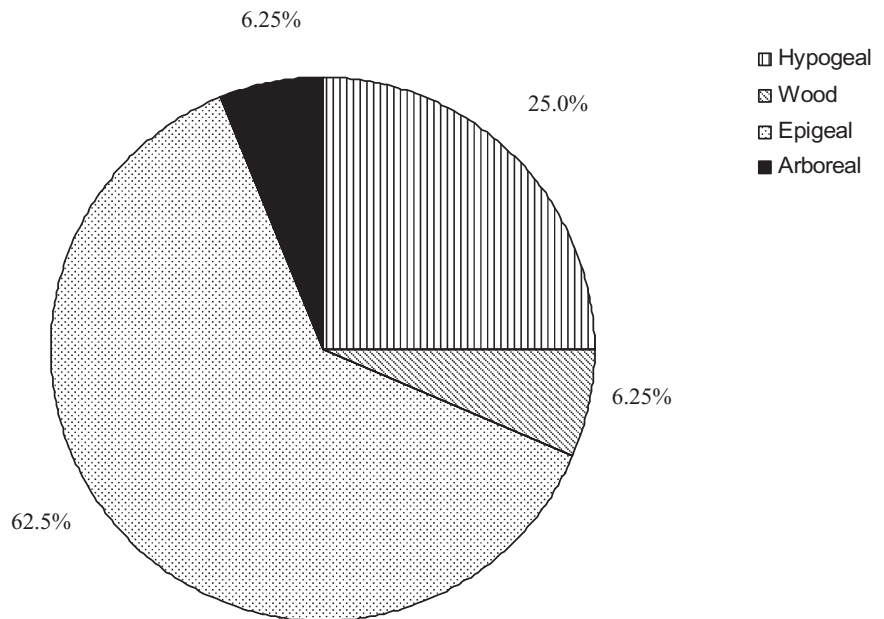


Fig. 4. Proportions of different nesting groups represented across the 16 termite species collected from direct search and bait sampling in Charles Darwin National Park, Darwin, Northern Territory.



whole appears to be typical for NT savanna woodland fauna in taxonomic and functional group structure. However, based on limited species inventory information available, I suggest it appears to be less species-rich by comparison; characterised by a lower relative abundance of soil- and grass-harvesting species.

The termite species inventory composed by Braithwaite *et al.* (1988) provides the only comprehensive comparison of termite assemblage structures for northern Australia. Braithwaite *et al.* (1988) recorded 50 species across five habitat types. The addition of bait sampling may have also further increased the number of species sampled (Dawes-Gromadzki 2005a). Monsoon forests were species-poor (23 species) compared with the surrounding open forest (36 species) and savanna woodland (35 species) habitats.

The CDNP termite community is relatively species-poor, at 16 species, compared with the savanna woodland sites (35

species) and even the species-poor monsoon forest sites (23 species) of Braithwaite *et al.* (1988). The lack of repeated/seasonal sampling and the fewer sites surveyed may contribute to the observed difference. Fourteen of the 16 species sampled in the present study were recorded in Kakadu savanna woodland by Braithwaite *et al.* (1988). The three most abundant species in the present study (*N. eucalypti*, *S. actuosus* and *M. boreus*) were also among the most common species encountered by Braithwaite *et al.* (1988). However, in the later study *M. darwiniensis* was more common and the grass harvester *Amitermes laurensis* was the most frequent species sampled.

The current study did support Braithwaite *et al.*'s (1988) finding that savanna woodland habitats are generally richer in species compared with nearby monsoon rainforest. Using the same methodology as the present study, Dawes-Gromadzki (2005a) recorded only five species in a monsoon rainforest

reserve located a few kilometres from CDNP. These Australian studies contrast to those from African savannas, where woodland habitats are generally species-poor compared with the species-rich rainforest habitats (Wood 1976; Dibog *et al.* 1999; Dawes-Gromadzki 2005a).

Baiting studies in other savanna woodland habitats near Darwin yielded a greater number of species than that recorded from baits in the present study. Dawes-Gromadzki and Spain (2003) recorded 11 species from five genera at paper baits and Dawes-Gromadzki (2003a,b) recorded 16 species from nine genera at paper and stake baits. However, the later studies were more intensive, increasing the chance of detecting rarer or seasonally active species.

A number of soil characteristics can also influence termite species diversity and abundance (Lee & Wood 1971; Jones & Prasetyo 2002; Dawes-Gromadzki 2005a). Soils with poor drainage capacity or inundation susceptibility can inhibit activity and provide unsuitable conditions for soil-dwelling and mound-building species (Jones 2000; Dawes-Gromadzki 2005a). In such environments, wood- and arboreal-nesting species are common. The dominance of mound- and soil-dwelling species in the present study is consistent with the elevated, well-draining kandosol soils of CDNP. However, although the species diversity observed is greater than that observed for seasonally inundated monsoon forest sites nearby (Dawes-Gromadzki 2005a), the relatively lower species diversity compared with Kakadu woodland may relate to soil fertility. In contrast to studies from tropical African and Indonesian rainforests (Dibog *et al.* 1999; Jones & Prasetyo 2002), Braithwaite *et al.* (1988) found a negative relationship between soil fertility and termite species richness. Compared with Kakadu, CDNP soils may be of generally lower fertility. At some sites soil depth appeared shallow, with patches of exposed rocky ground. This was particularly evident at site 4, which recorded the lowest species number. Patchiness in ground vegetation may also explain the lower species diversity (Braithwaite *et al.* 1988).

Termite feeding groups

The results of the current study only partially support the *a priori* predictions for termite feeding groups at CDNP, based on the work of Braithwaite *et al.* (1988). Wood and soil feeders were the dominant feeding groups in Kakadu savanna woodland, and grass-harvesting species were common (Braithwaite *et al.* 1988). In the present study wood feeders were dominant, but wood/soil interface feeders were also common. As predicted from Braithwaite *et al.*'s findings (1988), species from the wood-feeding genera *Mastotermes*, *Coptotermes*, *Schedorhinotermes* and *Microcerotermes* were also present at CDNP. The wood/soil interface feeder *N. eucalypti* dominated the CDNP termite fauna, followed by the dead wood feeders *S. actuosus* and *M. boreus*. Casual observations of termite communities around Darwin also support the dominance of wood and wood/soil-feeding termite species in savanna woodland (Dawes-Gromadzki & Spain 2003; G Ragno unpubl. data 2003; Dawes-Gromadzki 2005a).

In contrast to Braithwaite *et al.*'s (1988) findings where both live- and dead-wood feeding species were common, CDNP was mostly represented by species feeding on dead wood (*Schedorhinotermes* spp. and *Microcerotermes* spp.). The relatively lower soil nitrogen and phosphorus levels at CDNP may explain the relative absence of the live wood feeder *M. darwiniensis* as well as the particularly high frequency of occurrence of dead wood feeders in the present study compared with Kakadu (Braithwaite *et al.* 1988).

The absence of grass-harvesting and litter-feeding termites in CDNP does not support Braithwaite *et al.*'s (1988) findings in Kakadu savanna woodland where grass-harvesting species (*Amitermes*, *Tumulitermes* and *Nasutitermes* spp.) were common. At Kakadu, the dominant grass feeder was *A. laurensis*, whose NT distribution is confined to the western top end, and *Tumulitermes* spp. were also present. The only *Amitermes* spp. and *Nasutitermes* spp. recorded at CDNP were soil-feeding and wood/soil feeders, respectively. Grass feeding was only indirectly present through the polyphagous feeder *N. longipennis*.

The absence of a grass-feeding termite assemblage may reflect a low plant richness of the ground layer and general patchiness of ground vegetation. Braithwaite *et al.* (1988) found a negative relationship between the frequency of occurrence of harvester and soil-feeding species and ground layer plant richness. This would also explain the generally low occurrence of true soil feeders in the present study compared with the Kakadu woodland inventory.

Despite the above differences in feeding group assemblage structure, both these Australian studies do contrast to the generally greater termite diversity recorded for tropical savannas elsewhere. For example, the termite assemblages of African savannas are not only more diverse but are characterised by a significantly higher abundance of grass and grass-litter feeders (Wood *et al.* 1977; Ferrar 1982; Josens 1983; Braithwaite *et al.* 1988).

Termite nesting groups

The termite nesting habits documented at CDNP did support the *a priori* predictions based on the work of Braithwaite *et al.* (1988). At CDNP epigeal- and hypogeal-nesting species were dominant. Epigeal-nesting species were the most common followed by hypogeal-nesting species, supported by the high termite encounter rates in mounds and lying dead wood, respectively. In Kakadu, epigeal- and hypogeal-nesting species were also dominant. Wood- and arboreal-nesting species were also scarce, as in the present study. For the epigeal nesters, Braithwaite *et al.* (1988) also recorded species that inhabit the mounds of other termites, including *A. perelegans*.

Nasutitermes eucalypti was the dominant mound builder in the park, building small (cf. 0.5 m high) conical mounds each with a rounded apex. *Microcerotermes* spp. mounds are often built over dead stumps. Those of *M. boreus* are rounded, compared with the acutely pointed nests of *Microcerotermes nervosus*. *Microcerotermes serratus* builds small pointed or dome-shaped mounds or nests underground (Hill 1942;

Watson *et al.* 1998; Dawes-Gromadzki & Spain 2003). *Coptotermes acinaciformis* builds the largest mounds in the park; characteristic dome-shape structures often at the base of *Eucalyptus* trees (Dawes-Gromadzki & Spain 2003; Dawes-Gromadzki 2005a).

Both species of *Ephelotermes* build very small, black or grey earthy mound structures, usually at the base of a tree (often *E. melachoma*) or against lying dead wood (*Ephelotermes taylori*). *Ephelotermes taylori* may also nest underground, or in the mounds of other termites as does *A. perelegans* (Miller 1991; Watson *et al.* 1998). *Macrognathotermes* spp. mounds are black, earthy nest structures attached to existing mounds (Hill 1942; Miller 1991).

Schedorhinotermes actuosus nests underground or in wood buried in soil, and is often identified by the presence of thick, earthy sheeting on the surface of dead wood (Dawes-Gromadzki & Spain 2003; Dawes-Gromadzki 2005a). Both *Heterotermes* spp. form small colonies in soil, or in and under logs and stumps (Gay & Calaby 1970). *Mastotermes darwiniensis* was the only wood-nesting species collected, nesting in the trunks and branches of living trees, and standing and fallen dead wood (Watson *et al.* 1998; Dawes-Gromadzki & Spain 2003). *Nasutitermes graveolus* was the only arboreal nester, building spherical nests up in tree branches, with covered runways on the tree trunk connecting the nest to the ground (Dawes-Gromadzki & Spain 2003; Dawes-Gromadzki 2005a).

Biogeography of the Charles Darwin termite fauna

The documentation of *E. taylori* at CDNP represents the first record from open savanna woodland habitat, and the second record for the Darwin region; the first being from monsoon rainforest (Dawes-Gromadzki 2005a). *Ephelotermes taylori* was thought to be endemic to forest and rockland habitats in the 'top end' (restricted to the Kakadu region) of the NT (Braithwaite *et al.* 1988). However, these two later encounters now extend their geographic distribution west from Kakadu to the Darwin region.

The current study also represents the first record of *Macrognathotermes errorator* from savanna woodland. Casual collections of this species from rainforests indicate a wide distribution across the top end and Queensland (Qld) but there is an absence of official records (Watson & Abbey 1993).

The 14 other species in CDNP are typically found in open savanna woodland, as well as other habitats in the NT (Braithwaite *et al.* 1988; Dawes-Gromadzki & Spain 2003). *Nasutitermes eucalypti* and *N. longipennis* are both endemic to northern Australia; their distribution mostly is confined to northern NT, and for *N. longipennis* also northern Qld. The single record of *N. graveolus* reflects this species more common affinity for mostly coastal habitats across the NT and Qld (Braithwaite *et al.* 1988; Dawes-Gromadzki 2005a; ANIC online: <http://www.csiro.au/places/ANIC.html>).

Schedorhinotermes actuosus and *M. serratus* are widely distributed across Australia, and are common in savanna woodland and native forest NT habitats (Braithwaite *et al.*

1988; Dawes-Gromadzki 2003a,b; Dawes-Gromadzki & Spain 2003). *Microcerotermes boreus* and *M. nervosus* are mostly confined to northern NT, occupying a wide range of habitats (Watson & Abbey 1993). *Heterotermes venustus* occurs across northern Australia but is common towards the coast in native forest and savanna woodland. *Heterotermes vagus* occupies a range of habitat types but its distribution is restricted to the northern-most region of the NT and Qld (Watson & Abbey 1993; Dawes-Gromadzki 2003a).

Coptotermes acinaciformis and *M. darwiniensis* are endemic to mainland northern Australia, and occur widely throughout open forest, woodland and monsoon forest habitats (Dawes-Gromadzki & Spain 2003; Dawes-Gromadzki 2005a). The distribution of *Macrognathotermes sunteri* is tightly confined to north-east NT and the north tip of Western Australia (WA), but occurs in a range of habitats from monsoon forest to open woodland (Braithwaite *et al.* 1988; Watson & Abbey 1993). *Amitermes perelegans* and *E. melachoma* have been recorded in WA and Qld, but are most common to top end open forest and woodland habitats (Braithwaite *et al.* 1988; ANIC online).

In summary, I suggest that based on previous work some fairly accurate but limited predictions may be made for estimating the structure of savanna woodland termite assemblages in the NT. At present, however, these are severely restricted to broad estimations of the presence of common termite genera, species and functional groups, as comprehensive inventories of termite species diversity in Australia are scarce.

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